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March 3, 2026

Docket No.: 50-348 NL-26-0058

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Joseph M. Farley Nuclear Plant - Unit 1
Licensee Event Report 2026-001-00
Three Pressurizer Safety Valves' Lift Pressure Outside of Technical Specifications Limits due to Setpoint Drift

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B), Southern Nuclear Company is submitting the enclosed Licensee Event Report for Unit 1. The enclosed LER describes a condition which was prohibited by Technical Specifications.

This letter contains no NRC commitments. If you have any questions regarding this submittal, please contact Sarah Temple Licensing Project Manager, at (334) 661-3037.

Respectfully submitted,

Edwin Dean III

Vice President – Farley

ED/set/cbg

Enclosure: Unit 1 Licensee Event Report 2026-001-00

Cc: Regional Administrator, Region II
NRR Project Manager – Farley Nuclear Plant
Senior Resident Inspector – Farley Nuclear Plant
RTYPE: CFA04.054

Joseph M. Farley Nuclear Plant - Unit 1
Licensee Event Report 2026-001-00
Three Pressurizer Safety Valves' Lift Pressure Outside of Technical Specifications Limits due to Setpoint Drift

Enclosure

Unit 1 Licensee Event Report 2026-001-00

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name		2. Docket Number	3. Page
Joseph M. Farley Nuclear Plant, Unit 1	■ 050 ☐ 052	348	1 OF 2

4. Title: Three Pressurizer Safety Valves' Lift Pressure Outside of Technical Specifications Limits due to Setpoint Drift

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
01	06	2026	2026	- 001 -	00	03	03	2026	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Sarah Temple, Licensing Project Manager	(334) 661-3037

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	AB	RV	C710	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On January 6, 2026, while at 100% power and Mode 1, it was discovered that three Unit 1 pressurizer safety valves (PSVs), which had been removed during the fall 2025 refueling outage (1R33) and shipped off-site for testing, failed their as-found lift pressure tests. The three PSVs lifted below the Technical Specification (TS) 3.4.10 allowable lift setting value. Two of the PSVs also tested below the ASME code allowance of three percent. Setpoint drift is the cause of the PSV failures.

It is likely that the PSVs were outside of the TS limits longer than the allowable completion times for the associated required action statements during the previous operating cycle in all applicable modes of operation. Therefore, this condition is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by TS.

The PSVs were replaced during the fall 2025 refueling outage (1R33).

LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Joseph M. Farley Nuclear Plant, Unit 1	■ 050 ☐ 052	348	2026	- 001 -	00

NARRATIVE

EVENT DESCRIPTION:

During the Unit 1 fall 2025 refueling outage (1R33), the 1A, 1B, and 1C pressurizer safety valves (PSVs) [EIIS: RV] were removed for testing at an off-site testing facility and were replaced as part of an extent of condition from Unit 2 PSV failures (Unit 2 LER 2025-002-00). On January 6, 2026, while in Mode 1 and at 100% power, the site was notified that the as-found lift pressures of the 1A, 1B, and 1C PSVs were the following:

Valve Q1B13V031A (serial number N56963-01-0013) as found test pressure was determined to be 2397 psig (-3.5%).
Valve Q1B13V031B (serial number N56963-01-0012) as found test pressure was determined to be 2410 psig (-3.0%).
Valve Q1B13V031C (serial number N56963-01-0004) as found test pressure was determined to be 2416 psig (-2.8%).

These as-found lift pressures were discovered to be outside of the Technical Specification (TS) 3.4.10 allowable lift pressure settings of >/= 2423 psig and </= 2510 psig (-2.5% to +1%). The 1A and 1B PSVs were also found to be outside the ASME code acceptance band of +/- 3% (2410-2560 psig). No scope expansion was required per the ASME OM Code or the IST program as all valves in the group were tested.

EVENT ANALYSIS:

Setpoint drift was determined to be the cause of the PSV failures (Manufacturer: Crosby, Model Number: HP-BP-86, Serial Numbers: N56963-01-0004, N56963-01-0012, and N56963-01-0013).

REPORTABILITY AND SAFETY ASSESSMENT:

This failure constitutes a condition that is reportable pursuant to 10 CFR 50.73(a)(2)(i)(B), "Any operation or condition which was prohibited by the plant's Technical Specifications." There is no firm evidence, prior to the time of discovery at the test facility, of when the failure occurred.

None of the associated PSV as-found setpoints affected or challenged the ASME code limitation and safety limit of 110% of RCS design pressure. An engineering analysis was performed based on the as-found lift pressures and it was determined that the analysis done for the Unit 2 PSV failures (Unit 2 LER 2025-002-00) was bounding. There was no loss of safety function for this event.

CORRECTIVE ACTIONS:

All three PSVs were replaced during the fall 2025 refueling outage. Prior to installation, each valve was preconditioned to vendor recommendations and the as-left setpoints were within +/- 1% tolerance.

PREVIOUS SIMILAR EVENTS:

Similar events were reported over the past 3 years in Unit 2 LER 2025-002-00, Unit 1 LER 2024-001-00, and Unit 2 LER 2023-003-00.

OTHER SYSTEMS AFFECTED:

No other systems were affected by this event.